Optimum Post-Quantum Cryptographic Algorithms for a Monero Hardfork: A Comprehensive Architectural Blueprint

Executive Summary

The acceleration of quantum computing research poses a definitive and existential threat to the cryptographic foundations of privacy-preserving decentralized ledgers. Protocols such as Monero currently rely on the computational intractability of the Elliptic Curve Discrete Logarithm Problem to ensure transaction confidentiality, sender anonymity, and recipient unlinkability. The advent of cryptographically relevant quantum computers running Shor's algorithm will render these protections obsolete, allowing adversarial entities to retroactively de-anonymize the entire blockchain history, expose user balances, and forge spend authorizations to artificially inflate the currency supply. As the window for proactive mitigation narrows, engineering a post-quantum hardfork for the Monero network requires an intricate balance between robust quantum resistance and stringent operational constraints. Specifically, the proposed cryptographic suite must be lightweight and performant enough to be executed on consumer-grade hardware within a five-year horizon, while strictly avoiding exponential blockchain bloat.

This comprehensive report provides an exhaustive analysis of post-quantum cryptographic algorithms to determine the optimal architecture for a Monero hardfork. The findings indicate that a hybrid lattice-based architecture represents the only mathematically viable path forward. By integrating the Module Learning With Errors framework for stealth addresses, Lattice-based Aggregable Confidential Transactions for ring signatures, and the Lantern framework for zero-knowledge range proofs, the network can maintain its privacy guarantees. Coupled with Monero's upcoming Full-Chain Membership Proofs and the Carrot addressing protocol, this architectural blueprint ensures a highly scalable, computationally performant, and quantum-secure ledger designed to thrive over the next decade.

The Quantum Threat Landscape for Privacy Blockchains

The security of modern public-key cryptography rests fundamentally on asymmetric mathematical problems that are easy to compute in one direction but practically impossible to reverse using classical computational models. The most prominent of these are the Integer Factorization Problem and the Elliptic Curve Discrete Logarithm Problem.[1] Monero's entire cryptographic stack is built upon the latter, utilizing algorithms such as Ed25519 and secp256k1

for stealth addresses and transaction signing.[2]

Quantum computers operate on principles of quantum mechanics, utilizing qubits that can exist in a state of superposition, enabling them to process massive theoretical combinations simultaneously.[3] When a cryptographically relevant quantum machine achieves a sufficient number of stable, logical qubits, it can execute Shor's algorithm. This algorithm can solve the discrete logarithm problem in polynomial time, completely breaking elliptic curve cryptography.[5] For transparent ledgers like Bitcoin, this implies the potential for forged signatures and stolen funds.[7] However, for privacy-centric networks like Monero, the threat is exponentially more devastating. A quantum adversary would not only be able to steal funds but could completely unravel the privacy guarantees of the network, tracing the origin and destination of every transaction ever recorded.[3]

Furthermore, blockchains function as immutable, publicly distributed ledgers, making them highly susceptible to "harvest now, decrypt later" attack vectors.[1] Every Monero transaction broadcasted today is permanently recorded by network nodes across the globe. If the underlying cryptography is broken in the future, the privacy of today's transactions will be retroactively destroyed.[3] This creates an urgent, ticking clock for a post-quantum migration. The challenge, however, is that post-quantum cryptographic algorithms universally suffer from significantly larger key and signature sizes compared to their classical counterparts.[6] Naively implementing these algorithms into a blockchain protocol would precipitate explosive state growth, rendering the network un-verifiable for consumer-grade nodes and severely damaging the decentralization of the system.[8]

Deconstructing the Cryptographic Vulnerability Surface of the Monero Protocol

To architect a robust hardfork, it is necessary to deconstruct Monero's current cryptographic stack and identify the specific vulnerabilities embedded within its transaction lifecycle. Monero utilizes multiple interdependent cryptographic primitives to ensure sender anonymity, receiver unlinkability, and transaction value confidentiality. Every layer of this stack is currently reliant on elliptic curve cryptography and must be systematically replaced.

The Vulnerability of Stealth Addresses

To protect the identity of the recipient, Monero implements a Dual-Key Stealth Address Protocol.[2] When a sender initiates a transaction, they perform an elliptic curve Diffie-Hellman key exchange using the receiver's public view key and public spend key. This interaction generates a one-time ephemeral public key that becomes the destination address for the funds.[10] Because the mathematical derivation relies heavily on the hardness of the discrete logarithm problem over elliptic curves, a quantum computer could trivially derive the recipient's private keys from their publicly posted stealth meta-address.[5] By calculating the shared secret,

the adversary could link all historical and future ephemeral outputs to a single human entity, completely breaking receiver unlinkability.[5]

The Collapse of Ring Signatures and Spend Authorization

To obscure the true sender of a transaction, Monero utilizes ring signatures, specifically the Concise Linkable Spontaneous Anonymous Group scheme. This cryptographic mechanism proves that one member out of a group of network participants authorized the transaction without revealing which specific member produced the signature.[2] The protocol includes a "key image" designed to prevent double-spending by mathematically linking the signature to the spent output without revealing the output itself.[14] Because the anonymity of the ring signature is entirely derived from the discrete logarithm assumption, a quantum adversary could compute the discrete logarithm of the key image and the signature components.[2] This computation would instantly unmask the true signer among the decoys, destroying sender privacy and allowing the adversary to trace the exact flow of funds across the network.[2]

The Failure of Ring Confidential Transactions and Bulletproofs

Monero hides the exact amount of currency transferred in a transaction using Pedersen commitments. A Pedersen commitment is a homomorphic cryptographic primitive that allows the network to verify that the inputs of a transaction equal the outputs (plus the transaction fee) without ever revealing the actual amounts.[2] However, because the commitments are homomorphic, an attacker could theoretically create a transaction with a negative output value, thereby creating infinite coins out of thin air. To prevent this, Monero uses Bulletproofs, a zero-knowledge range proof protocol that cryptographically guarantees that the committed values lie within a positive integer range.[2]

Bulletproofs are highly favored in classical cryptography because they are incredibly compact, typically requiring less than one kilobyte of data.[16] Unfortunately, they are entirely reliant on the discrete logarithm assumption.[2] Breaking the underlying assumptions of the Ring Confidential Transactions protocol would allow a quantum attacker to view all historical transaction amounts in cleartext. Furthermore, through algebraic manipulation of the broken commitments, the attacker could easily identify the true inputs in a ring signature, rendering the decoy mechanisms useless.[5]

Comprehensive Evaluation of Post-Quantum Cryptographic Families

The cryptographic community and the National Institute of Standards and Technology have standardized several post-quantum algorithms designed to resist Shor's algorithm. However, blockchain environments impose unique constraints that differ vastly from traditional secure web communications. Algorithms deployed on a blockchain must minimize on-chain byte size

to prevent the ledger from outgrowing consumer storage, feature rapid verification times to maintain node synchronization limits, and avoid excessive memory consumption during transaction validation.[8] An exhaustive evaluation of the primary post-quantum families yields critical insights for the hardfork architecture.

Hash-Based Cryptography

Hash-based signature schemes, such as SPHINCS+ and the eXtended Merkle Signature Scheme, rely entirely on the proven security of cryptographic hash functions.[6] Because they do not rely on complex mathematical assumptions like lattices or isogenies, they offer the most conservative and highly trusted security guarantees against quantum attacks.[6] However, they are fundamentally unsuited for a high-frequency blockchain network like Monero. Hash-based signatures are notoriously massive. A standard SPHINCS+ signature can require multiple kilobytes of data, often exceeding eight to thirty kilobytes depending on the specific parameter set utilized for security.[6] Implementing such a scheme would cause exponential blockchain bloat, rapidly pushing the ledger size into the terabytes and centralizing node operation.[6] Furthermore, stateful schemes require meticulous tracking of one-time key usage; in a decentralized, asynchronous network, state synchronization failures can lead to key reuse and catastrophic security breaches.[6]

Multivariate Cryptography

Multivariate public key cryptography relies on the difficulty of solving large systems of multivariate polynomial equations over finite fields.[20] While these schemes can theoretically produce very short signatures, they have historically suffered from structural fragilities, complex parameter tuning, and exceptionally large public key sizes.[20] The catastrophic algorithmic break of the Rainbow signature scheme by classical computers during the standardization process underscores the immaturity and high risk associated with deploying multivariate cryptography in a high-stakes financial ledger.[6]

Isogeny-Based Cryptography

Isogeny-based cryptography, represented by protocols such as the Commutative Supersingular Isogeny Diffie-Hellman scheme and SQIsign, constructs cryptographic protocols based on pseudorandom walks through supersingular elliptic curve isogeny graphs.[11] This family is highly attractive because it offers the absolute smallest key and signature sizes in the post-quantum landscape, often maintaining byte footprints comparable to classical elliptic curve cryptography.[11] For instance, the Erebor variant of SQIsign yields a signature size of just 0.71 kilobytes for an anonymity set of four users.[23] However, the computational overhead of isogeny mathematics is a massive operational bottleneck. Isogeny operations are orders of magnitude slower than lattice or hash-based schemes, requiring highly intensive mathematical computations that would severely cripple block verification times and node synchronization

speeds on standard consumer hardware.[23]

Lattice-Based Cryptography

Lattice-based cryptography represents the optimal compromise between provable security, signature size, and computational efficiency.[6] These systems are built upon hard mathematical problems such as Learning With Errors, Ring Learning With Errors, Module Learning With Errors, and the Short Integer Solution problem.[10] The primary standardized algorithms, including the Key Encapsulation Mechanism Kyber and the Digital Signature Algorithm Dilithium, belong to this family.[6]

While lattice signatures are indeed larger than classical elliptic curve signatures—with Dilithium averaging between 2.4 to 4.6 kilobytes—their verification speeds are exceptionally fast, and they allow for advanced cryptographic constructs like zero-knowledge proofs and homomorphic commitments.[6] The evaluation indicates that while lattice-based schemes will increase the base size of a transaction, their algorithmic flexibility allows for architectural optimizations (such as transaction aggregation and inner-product arguments) that can dramatically mitigate blockchain bloat. For a projected five-year implementation timeline, hardware acceleration and advanced vector extensions will fully neutralize the computational overhead of lattice mathematics, making this family the definitive and optimum choice for Monero's hardfork.[25]

Engineering Post-Quantum Stealth Addresses

Replacing the classical Dual-Key Stealth Address Protocol requires a Key Encapsulation Mechanism that provides indistinguishability under chosen-ciphertext attack. The optimal solution identified through this research is the implementation of the Module Learning With Errors Stealth Address Protocol.[10]

The Module-LWE Stealth Address Architecture

Module-LWE balances the strict, conservative security guarantees of standard Learning With Errors with the high operational efficiency of Ring Learning With Errors by defining lattices over structured module matrices.[24] In the proposed protocol architecture, the recipient generates a stealth meta-address consisting of two distinct public keys derived via the Kyber key generation function.[24] When a sender wishes to transmit funds to this recipient, they execute the encapsulation algorithm using the recipient's public view key to generate an ephemeral ciphertext alongside a cryptographically secure shared secret.[24] The sender then uses this shared secret to deterministically calculate the specific ephemeral transaction address.

The performance profile of the Module-LWE protocol is vastly superior to other post-quantum proposals in the literature. Empirical benchmarks and implementation analyses demonstrate that the Module-LWE approach achieves a 66.8% faster scanning time compared to the

classical Elliptic Curve Pairing Dual-Key protocol, specifically outperforming the highly efficient Curvy protocol designed for Ethereum compatibility.[10] When evaluated across varying announcement loads, scaling up to 80,000 ephemeral keys, the Module-LWE implementation maintains a highly linear and efficient scan time, effectively matching the speed of classical hybrid protocols while offering full quantum resistance.[12]

KEM Parameter Set	Estimated Time (ms) per 5000 Scans	Security Level Equivalency
Kyber512	87	AES-128
Kyber768	141	AES-192
Kyber1024	205	AES-256
NewHope1024	293	AES-256
Frodo1344	24,806	AES-256

Table 1: Scan time performance across various Key Encapsulation Mechanisms utilizing a one-byte view tag. Kyber768 represents the optimum balance of post-quantum security and scanning latency.[10]

Registry Optimization via Cryptographic View Tags

A critical structural bottleneck in any stealth address protocol is the computational burden placed on the receiver. To discover incoming funds, a wallet must cryptographically scan every single transaction broadcasted to the blockchain. In a lattice-based paradigm, running a full Kyber decapsulation routine for every network transaction is computationally prohibitive, particularly for lightweight wallets running on mobile devices.[12]

To optimize this process for a realistic 2031 deployment, the hardfork architecture must implement "view tags." A view tag is a highly truncated cryptographic hash of the shared secret, appended directly to the transaction data payload.[12] During the scanning process, the receiver's wallet calculates a preliminary hash. If the transaction does not belong to the user, this one-byte view tag filters out 99.6% of transactions instantly without requiring further mathematical operations.[12] The recipient only performs the heavy Module-LWE decapsulation if the view tag matches their locally computed expectation.[12] Extensive analysis confirms that a one-byte view tag represents the optimal operational trade-off: it adds a negligible single byte of data per transaction while accelerating blockchain scanning by two full orders of magnitude,

making post-quantum stealth addresses perfectly viable for mobile hardware.[10]

Quantum-Resistant Ring Signatures and Spend Authorization

The core of Monero's privacy model relies entirely on hiding the true sender among a group of decoys. Transitioning this function to a post-quantum framework is arguably the most challenging aspect of the hardfork due to the extreme risk of blockchain data bloat.

To enhance privacy, the Monero Research Lab is currently transitioning the network toward Full-Chain Membership Proofs.[31] This upgrade abandons traditional ring signatures, which typically utilize a limited anonymity set of sixteen decoys, in favor of proving membership against the entire set of historical outputs recorded on the ledger.[31] This effectively upgrades the decoy count to over one hundred million. However, the current iteration of Full-Chain Membership Proofs is designed around Curve Trees, which operate on elliptic curves and are fundamentally vulnerable to quantum algorithms.[14] Therefore, any chosen post-quantum signature scheme must possess a proof size that scales logarithmically, rather than linearly, with the size of the anonymity set. If the proof size scales linearly, referencing millions of decoys would result in transactions measured in gigabytes.[32]

The Evolution of Lattice Proofs: From MatRiCT to MatRiCT+

The MatRiCT protocol emerged as the first practical lattice-based Ring Confidential Transaction scheme, providing a foundational blueprint for developing post-quantum privacy coins.[17] It successfully implemented lattice-based balance proofs with hashed-message commitments. However, the original MatRiCT protocol suffered from severe inefficiencies. Its proof sizes scaled linearly with the number of transaction inputs, making it wholly unsuitable for complex multi-input transactions or massive anonymity sets.[32] Furthermore, it required the generation and verification of complex "corrector values" during binary proofs, which imposed a massive computational penalty on the system.[17]

The enhanced MatRiCT+ protocol resolves these architectural deficiencies by utilizing the specific algebraic properties of power-of-two cyclotomic rings, combined with a novel "partition-and-sample" technique for generating challenge spaces.[32] This drastically minimizes the mathematical norm of the challenges, allowing for highly compact zero-knowledge proof systems. The architectural improvements achieved by MatRiCT+ are profound:

- **Proof Size Compression:** By leveraging unbalanced linear sum approaches and avoiding costly binary proofs, MatRiCT+ generates proofs that are 2 to 18 times shorter than the original protocol, scaling logarithmically rather than linearly.[17]
- **Execution Velocity:** Due to highly optimized rejection sampling algorithms and inner-product proofs, MatRiCT+ runs 3 to 11 times faster for standard cryptocurrency transactions.[32]

- **Corrector Value Elimination:** Advanced lattice-based zero-knowledge proofs further optimize the protocol by completely removing the need for corrector values, reducing the required proving time by up to 30% and verification time by 20%, while shrinking the proof size by an additional 15% to 20% compared to baseline metrics.[17]

Despite these massive optimizations, MatRiCT+ still yields a relatively high per-transaction byte footprint. For an anonymity set of 64 members, standard lattice-based group signatures can range from 89 kilobytes to 126 kilobytes depending on the strictness of the security parameters.[23] If deployed universally across the network without further mitigation, the continuous accumulation of these signatures would eventually violate the constraint against a burdensome blockchain length.

Preventing Blockchain Bloat: Lattice-Based Aggregable Confidential Transactions

To satisfy the stringent requirement of ensuring long-term ledger viability over the next decade, the Monero hardfork must look beyond traditional, static transaction models and adopt Lattice-Based Aggregable Confidential Transactions.[34]

Standard confidential transaction protocols, including the highly optimized MatRiCT+, require every single transaction to permanently record its mathematical commitments, ring signatures, and range proofs on the blockchain data structure. This data is rigorously immutable and eternally expands the physical size of the ledger, a phenomenon heavily criticized as an unexamined structural flaw in modern privacy blockchains.[8] Lattice-Based Aggregable Confidential Transactions introduces a radical paradigm shift in ledger management: it allows multiple individual transactions to be mathematically aggregated into a single, highly compressed transaction block.[35] Through this aggregation mechanism, the network can permanently delete intermediate spent coin records and redundant signature data while maintaining cryptographically trustless, consensus-level verification.[35]

The profound efficiency gains of this aggregable protocol are heavily dependent on its utilization of the Approximate Short Integer Solution problem and the implementation of logarithmic carry proofs.[36]

- **Commitment Compression:** By strictly leveraging the properties of the Approximate Short Integer Solution, the protocol reduces the size of confidential monetary value commitments from the 9.6 kilobytes required by MatRiCT+ down to a highly optimized 5.7 kilobytes.[36]
- **Header and Proof Footprint:** A complete, unaggregated transaction header, which contains the digital signature, the logarithmic carry proof, and the activity proof, occupies approximately 10.3 kilobytes of space.[36]
- **Aggregable Activity Proofs:** The absolute most crucial feature of this protocol is its mechanism for maintaining consensus-level immutability. It achieves this via an activity

proof that is statically sized at exactly 49 bytes, regardless of whether the transaction has two inputs or two thousand inputs.[35] This constant-size proof does not break the aggregation mechanics. Consequently, network nodes can download a heavily pruned, fully aggregated chain and still mathematically verify the existence and integrity of all historical balances without blindly trusting third-party peers.[35]

Cryptographic Protocol	Base Mathematical Assumption	Commitment Size	Proof Scaling Metric	Transaction Aggregability	Verification Latency
Monero CLSAG	Elliptic Curve Discrete Log	< 1 KB	Linear	None	Fast (Vulnerable)
MatRiCT	Lattice (Module-SIS)	4.48 KB	Linear	None	Moderate
MatRiCT+	Lattice (Cyclotomic Rings)	9.6 KB	Logarithmic	None	Fast
LACT+	Lattice (Approx-SIS)	5.7 KB	Logarithmic	High (Consensus)	Very Fast

Table 2: Comparative architectural analysis of Ring Confidential Transaction models. The LACT+ protocol represents the optimal intersection of data footprint and operational scalability for a post-quantum Monero network.[32]

By implementing this aggregable architecture, Monero can successfully decouple its daily transaction volume from its permanent storage requirements. Even though the baseline lattice-based proofs are natively larger than historical elliptic curve proofs, the continuous, mathematically sound pruning of spent outputs ensures that the long-term storage growth of the blockchain is fundamentally flatter than a classical, non-aggregable chain.[36]

Overcoming the Zero-Knowledge Range Proof

Bottleneck

Monero currently utilizes the Bulletproofs algorithm to cryptographically prove that the hidden amounts transferred in a transaction represent positive integers.[2] This is critical to prevent the creation of illicit funds. Bulletproofs are exceedingly efficient, requiring less than one kilobyte of data space, but their security relies entirely on the discrete logarithm assumption and folding strategies that are fundamentally incompatible with post-quantum security models.[2] Replacing Bulletproofs with a quantum-resistant equivalent is a severe technical hurdle because lattice-based zero-knowledge range proofs are notoriously complex and bulky.

Early attempts at constructing lattice-based range proofs required tens or hundreds of kilobytes, relying on repetitive Stern-like protocols that drastically increased communication costs and made them unviable for high-throughput cryptocurrency applications.[38] Alternative approaches, such as the ALS protocol, utilized inner product arguments and n-ary decomposition to reduce sizes, but their proofs could not be seamlessly aggregated.[38]

The Lantern Zero-Knowledge Framework

The recently developed "Lantern" framework provides a highly optimized lattice-based zero-knowledge argument protocol that compresses proof sizes down to practical, real-world margins.[37] Lantern operates securely on the established hardness of the Module-Short Integer Solution and Module-Learning With Errors problems.[37]

Traditional lattice proofs require converting values into a binary Chinese Remainder Theorem representation and explicitly proving that each individual coefficient of a vector is small, which requires massive, multi-dimensional matrices.[37] The Lantern framework bypasses this inefficiency entirely by utilizing sophisticated polynomial product proof systems.[37] By mathematically isolating the inner product of two secret vectors into the constant coefficient of a generated polynomial product, and strictly hiding all other coefficients, Lantern can prove knowledge of an inner product modulo the prime q directly and efficiently.[37]

Through an innovative "approximate range proof" mechanism, the Lantern framework effectively lifts the mathematical proof over the standard integers instead of the finite field modulus. This brilliant algorithmic breakthrough reduces the resulting proof size for a standard 64-bit transaction value to approximately 13 kilobytes.[37] While this represents a size increase over classical Bulletproofs, it is a staggering two to three times smaller than all preceding lattice-based range proof methodologies.[37] When seamlessly integrated into the transaction aggregation structures of the LACT+ protocol, the 13-kilobyte footprint of the Lantern proof is efficiently amortized across multiple transaction inputs and outputs, completely neutralizing its impact on the long-term size of the ledger.

The 5-Year Hardware Horizon: Computational

Performance by 2031

A primary concern raised by critics of lattice-based cryptography is its high computational overhead. The mathematical operations required—specifically the Number Theoretic Transform, complex polynomial multiplication, and constant-time rejection sampling—are significantly more demanding on a processor than traditional elliptic curve scalar multiplication.[25] However, projecting hardware and silicon capabilities five years into the future reveals that these temporary computational bottlenecks will be entirely mitigated by the architecture of consumer-grade devices.

Advanced Vector Extensions and SIMD Processing

The dominant computational bottleneck in lattice cryptography is polynomial convolution.[25] While the Number Theoretic Transform reduces the algorithmic complexity of this operation, it remains a heavy burden on standard processors.[25] To address this, modern CPU architectures are aggressively integrating dedicated instruction sets for Single Instruction, Multiple Data parallel processing. The widespread adoption of Advanced Vector Extensions on x86 architectures, alongside the integration of the Scalable Vector Extension 2 on the ARMv9 architecture, provides massive hardware-level parallelization for polynomial operations.[27]

Extensive benchmarks of the Dilithium and Kyber algorithms executed on ARM Cortex-A72 and Apple M1 processors utilizing NEON vector instructions demonstrate immense speed improvements. By implementing interleaved multi-stage butterflies combined with novel "Barrett multiplication"—a technique fusing Montgomery multiplication and Barrett reduction—researchers have achieved performance speeds 1.7 to 2.1 times faster than previous state-of-the-art software implementations.[39]

By the target deployment year of 2031, ARMv9 processors with native Scalable Vector Extension 2 support will be ubiquitous across mobile devices, tablets, and consumer laptops.[27] Consequently, the verification latency currently observed for complex lattice proofs will be reduced from tens of milliseconds down to sub-millisecond ranges, ensuring that low-power nodes can synchronize the post-quantum Monero blockchain as fast as, or significantly faster than, the current elliptic curve implementation.[41]

Hardware-Accelerated Hash Cores

Lattice schemes heavily depend on Extendable-Output Functions, specifically the SHAKE-128 and SHAKE-256 algorithms, to deterministically expand tiny public seeds into massive, multi-dimensional matrices on the fly, thereby saving tremendous network bandwidth.[25] In early hardware implementations, the overall throughput of polynomial generation was frequently bounded by the clock speed of the hash core.[25] Over the next five years, hardware-accelerated SHA-3 and SHAKE instructions will become standard, integrated features on system-on-chip designs, entirely eliminating this bottleneck.[25] The powerful

combination of advanced vector extensions for the Number Theoretic Transform and dedicated hardware hash cores for rapid matrix generation guarantees that the transition to lattice-based aggregable transactions and Module-LWE stealth addresses will be seamlessly performant for the end-user.

Migration Strategy and Consensus Hardfork Implementation

Deploying a post-quantum hardfork is not merely a theoretical cryptographic challenge; it is an immense protocol engineering and decentralized consensus orchestration task. The transition protocol must protect users who are offline during the network fork while simultaneously ensuring that compromised elliptic curve keys cannot be used by a quantum adversary to silently hyper-inflate the monetary supply.

Integration with Full-Chain Membership Proofs

The Monero Research Lab has already begun the fundamental groundwork for this transition by architecting the network upgrade to Full-Chain Membership Proofs.[31] The current iteration of this upgrade relies on classical cryptography and is intended to be launched on an alpha stressnet to test node capabilities.[43] To prepare for the post-quantum era, the cryptographic accumulator used within these membership proofs must be seamlessly swapped with a lattice-based logarithmic accumulator derived directly from the principles of MatRiCT+ and the aggregable confidential transactions protocol.[14] Because the chosen lattice protocol scales logarithmically, it inherently supports the massive, whole-chain anonymity set required by the membership proofs without causing transaction sizes to mathematically explode.[32]

The Carrot Addressing Protocol and Forward Secrecy

To facilitate the transition to quantum-secure addresses without degrading the established user experience or requiring immediate wallet software rewrites, Monero is implementing the Carrot protocol.[44] Carrot introduces a highly flexible, novel key hierarchy that provides unconditional forward secrecy for change outputs and is mathematically designed to be indistinguishable on-chain from other proposed addressing schemes.[44]

Crucially for the post-quantum transition, the Carrot protocol introduces a mechanism known as "quantum-resistant churning".[13] Churning is the process by which a user sends funds to themselves to refresh their transaction history and obfuscate the origin of their currency. Under the Carrot architecture, these self-send transactions are heavily encrypted such that they cannot be decrypted or analyzed by a quantum computer relying purely on scraped on-chain data, provided the adversary has not physically compromised the user's public address hardware.[13]

The Migration Window and Switch Commitments

The absolute most sensitive and critical phase of the hardfork is the migration window. If a cryptographically relevant quantum computer suddenly comes online, attackers could instantly forge signatures for old, dormant, pre-fork outputs that are still secured by elliptic curve mathematics.[45] To counter this catastrophic threat, the protocol will utilize "switch commitments." This advanced cryptographic technique allows for the transparent revealing of transaction amounts during the specific migration phase, effectively neutralizing the risk of hidden inflation caused by a quantum attacker minting fake coins out of thin air.[13]

The strategic roadmap outlines a mandatory, network-enforced migration window, currently projected to be twelve months in duration.[46] During this specific epoch, users must execute a standard transaction to migrate their funds from classical elliptic curve outputs to the new, secure Module-LWE outputs. If a user fails to migrate their funds within this critical window, their funds are securely locked by the consensus rules. Spending these locked funds in the future will require the generation of a highly complex, computationally heavy proof of ownership specifically designed to withstand advanced quantum forgery attempts.[46] This forced migration architecture ensures that the active unspent transaction output set becomes strictly post-quantum, permanently quarantining legacy outputs that are susceptible to Shor's algorithm and securing the economic integrity of the protocol.

Architectural Synthesis and Conclusion

The impending transition to post-quantum cryptography represents the most formidable engineering, mathematical, and consensus challenge in the history of privacy-preserving decentralized ledgers. The dual threats of Shor's algorithm breaking the discrete logarithm problem and the persistent, invisible risk of "harvest now, decrypt later" surveillance dictate that the network migration must be proactive, comprehensive, and architecturally flawless.

By strategically abandoning classical elliptic curve cryptography and integrating the Module Learning With Errors framework for stealth addressing, the network secures recipient privacy against chosen-ciphertext attacks while maintaining lightning-fast wallet synchronization through one-byte view tags. By implementing the Lantern zero-knowledge framework, the network replaces vulnerable Bulletproofs with highly compressed, thirteen-kilobyte lattice range proofs that avoid the computational bloat of traditional Chinese Remainder Theorem representations.

Most importantly, to satisfy the critical operational constraint of preventing a burdensome blockchain length over the next decade, the Monero protocol must implement Lattice-Based Aggregable Confidential Transactions. This inherently aggregable architecture uniquely solves the primary historical drawback of lattice-based cryptography—data bloat—by empowering the network to continually compress and prune spent transaction histories down to mathematically verifiable, forty-nine-byte activity proofs. Anticipating the widespread

maturation and deployment of ARMv9, Scalable Vector Extension 2, and AVX-512 hardware by 2031, this meticulously designed hybrid lattice architecture guarantees that the Monero network will remain lightweight, computationally performant, and cryptographically impenetrable in the post-quantum era.

Works cited

1. Post-quantum cryptography - Wikipedia, accessed February 21, 2026, https://en.wikipedia.org/wiki/Post-quantum_cryptography
2. An SOK of How Post-Quantum Attackers Reshape Blockchain Security and Performance, accessed February 21, 2026, https://arxiv.org/html/2512.13333v1
3. Enhancing Privacy and Usability in Quantum-Resistant Cryptocurrencies - DigitalCommons@SHU, accessed February 21, 2026, https://digitalcommons.sacredheart.edu/cgi/viewcontent.cgi?article=2563&context=acadfest
4. dilithion/website/POST-QUANTUM-CRYPTO-COURSE.md at main - GitHub, accessed February 21, 2026, https://github.com/WillBarton888/dilithion/blob/main/website/POST-QUANTUM-CRYPTO-COURSE.md
5. Is Monero Quantum Secure? Explore the Leading Privacy Coin, accessed February 21, 2026, https://www.quantumcanary.org/insights/is-monero-quantum-secure-explore-the-leading-privacy-coin
6. Post-Quantum Roadmaps for Blockchain Ecosystems | by Jung-Hua ..., accessed February 21, 2026, https://medium.com/@gwrx2005/post-quantum-roadmaps-for-blockchain-ecosystems-af9e77a6fe8b
7. Will Quantum Computing Kill Bitcoin? - CoinGecko, accessed February 21, 2026, https://www.coingecko.com/learn/quantum-computing-bitcoin
8. Blockchains Don't Have a Privacy Problem. They Have a Memory Problem, accessed February 21, 2026, https://hackernoon.com/blockchains-dont-have-a-privacy-problem-they-have-a-memory-problem
9. Quantum Disruption: An SOK of How Post-Quantum Attackers Reshape Blockchain Security and Performance - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/398720912_Quantum_Disruption_An_SOK_of_How_Post-Quantum_Attackers_Reshape_Blockchain_Security_and_Performance
10. Post-Quantum Stealth Address Protocols - arXiv, accessed February 21, 2026, https://arxiv.org/html/2501.13733v1
11. Towards practical post quantum stealth addresses - Cryptography - Ethereum Research, accessed February 21, 2026, https://ethresear.ch/t/towards-practical-post-quantum-stealth-addresses/15437
12. (PDF) Post-Quantum Stealth Address Protocols - ResearchGate, accessed

February 21, 2026, https://www.researchgate.net/publication/388354033_Post-Quantum_Stealth_Address_Protocols
13. Discussion: Post-quantum security and ethical considerations over elliptic curve cryptography · Issue #131 · monero-project/research-lab - GitHub, accessed February 21, 2026, https://github.com/monero-project/research-lab/issues/131
14. Logarithmic-size signature LS-LSAG : r/Monero - Reddit, accessed February 21, 2026, https://www.reddit.com/r/Monero/comments/1ef7i2e/logarithmicsize_signature_lslsag/
15. Monero Becomes Bulletproof. A complete look at Monero's current... | by Lucas Nuzzi | Digital Asset Research | Medium, accessed February 21, 2026, https://medium.com/digitalassetresearch/monero-becomes-bulletproof-f98c6408babf
16. Bulletproofs and Rangeproofs - ASecuritySite.com, accessed February 21, 2026, https://asecuritysite.com/bulletproof
17. Lattice-based Zero-knowledge Proofs for Blockchain ... - PolyU, accessed February 21, 2026, https://www4.comp.polyu.edu.hk/~shanggao//publications/Lattice-based_Zero-knowledge_Proofs_for_Blockchain_Confidential_Transactions.pdf
18. Comprehensive Comparison of Post-Quantum Digital Signature Schemes in Blockchain, accessed February 21, 2026, https://ieeexplore.ieee.org/document/9686427/
19. Post-Quantum Digital Signatures for Blockchain - Dora Research Blog - DoraHacks, accessed February 21, 2026, https://research.dorahacks.io/2024/07/31/post-quantum-digital-signatures-for-blockchain/
20. Comparative Evaluation of Lattice-Based, Hash-Based, and Multivariate Post-Quantum Cryptographic Schemes | Request PDF - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/396970226_Comparative_Evaluation_of_Lattice-Based_Hash-Based_and_Multivariate_Post-Quantum_Cryptographic_Schemes
21. A Survey and Comparison of Post-quantum and Quantum Blockchains - arXiv, accessed February 21, 2026, https://arxiv.org/pdf/2409.01358
22. Post-Quantum Cryptography and Quantum-Safe Security: A Comprehensive Survey - arXiv, accessed February 21, 2026, https://arxiv.org/html/2510.10436v1
23. MatRiCT + : More Efficient Post-Quantum Private Blockchain Payments | Request PDF, accessed February 21, 2026, https://www.researchgate.net/publication/362297514_MatRiCT_More_Efficient_Post-Quantum_Private_Blockchain_Payments
24. Post-Quantum Stealth Address Protocols - arXiv.org, accessed February 21, 2026, https://arxiv.org/pdf/2501.13733
25. Lattice-Based Cryptographic Accelerators for the Post-Quantum Era ..., accessed February 21, 2026, https://www.mdpi.com/2079-9292/15/2/475

26. A Performance Comparison of Post-Quantum Algorithms in Blockchain, accessed February 21, 2026, https://jbba.scholasticahq.com/article/38508-a-performance-comparison-of-post-quantum-algorithms-in-blockchain/attachment/100487.pdf
27. [PDF] Optimized Vectorization Implementation of CRYSTALS-Dilithium | Semantic Scholar, accessed February 21, 2026, https://www.semanticscholar.org/paper/Optimized-Vectorization-Implementation-of-Zheng-Zhu/2633c8db5c983a4314e06ce487c6435d11b3e83e
28. A lattice-based linkable ring signature supporting stealth addresses - InK@SMU.edu.sg, accessed February 21, 2026, https://ink.library.smu.edu.sg/context/sis_research/article/8416/viewcontent/A_Lattice_Based_Linkable_Ring_Signature_Supporting_Stealth_Addresses.pdf
29. More Efficient Stealth Address Protocol, accessed February 21, 2026, https://arxiv.org/abs/2504.06744
30. A Comprehensive Survey on Pre Versus Post Quantum Security Schemes for 5G-Enabled IoT Applications - IEEE Xplore, accessed February 21, 2026, https://ieeexplore.ieee.org/iel8/6287639/10820123/11155090.pdf
31. Full-Chain Membership Proofs Development | Monero - secure, private, untraceable, accessed February 21, 2026, https://www.getmonero.org/2024/04/27/fcmps.html
32. MatRiCT>+>: more efficient post-quantum private blockchain payments - Monash University, accessed February 21, 2026, https://research.monash.edu/en/publications/matrictsupsup-more-efficient-post-quantum-private-blockchain-paym/
33. MatRiCT: Efficient, Scalable and Post-Quantum Blockchain Confidential Transactions Protocol | Request PDF - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/337099516_MatRiCT_Efficient_Scalable_and_Post-Quantum_Blockchain_Confidential_Transactions_Protocol
34. (PDF) LACT+: Practical Post-Quantum Scalable Confidential Transactions - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/370647382_LACT_Practical_Post-Quantum_Scalable_Confidential_Transactions
35. LACT+: Practical Post-Quantum Scalable Confidential Transactions - MDPI, accessed February 21, 2026, https://www.mdpi.com/2410-387x/7/2/24
36. LACT+: Practical Post-Quantum Scalable Confidential Transactions, accessed February 21, 2026, https://www.mdpi.com/2410-387X/7/2/24
37. Lattice-Based Zero-Knowledge Proofs Under a Few Dozen Kilobytes - ETH Zurich Research Collection, accessed February 21, 2026, https://www.research-collection.ethz.ch/bitstreams/79504434-0622-471a-b604-01870540b2fc/download
38. SoK: Zero-Knowledge Range Proofs - DROPS, accessed February 21, 2026, https://drops.dagstuhl.de/storage/00lipics/lipics-vol316-aft2024/LIPIcs.AFT.2024.14/LIPIcs.AFT.2024.14.pdf
39. Neon NTT: Faster Dilithium, Kyber, and Saber on Cortex-A72 and Apple M1, accessed February 21, 2026, https://d-nb.info/1274959330/34

40. Neon NTT: Faster Dilithium, Kyber, and Saber on Cortex-A72 and Apple M1 - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/356399114_Neon_NTT_Faster_Dilithium_Kyber_and_Saber_on_Cortex-A72_and_Apple_M1
41. Constrained Device Performance Benchmarking with the Implementation of Post-Quantum Cryptography - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/380846689_Constrained_Device_Performance_Benchmarking_with_the_Implementation_of_Post-Quantum_Cryptography
42. (PDF) Lattice-Based Cryptographic Accelerators for the Post-Quantum Era: Architectures, Optimizations, and Implementation Challenges - ResearchGate, accessed February 21, 2026, https://www.researchgate.net/publication/400038783_Lattice-Based_Cryptographic_Accelerators_for_the_Post-Quantum_Era_Architectures_Optimizations_and_Implementation_Challenges
43. When can we expect FCMP++ to be implemented? : r/Monero - Reddit, accessed February 21, 2026, https://www.reddit.com/r/Monero/comments/1p7xqja/when_can_we_expect_fcmp_to_be_implemented/
44. carrot/carrot.md at master · jeffro256/carrot · GitHub, accessed February 21, 2026, https://github.com/jeffro256/carrot/blob/master/carrot.md
45. Monero Research Lab Meeting - Wed 18 December 2024, 17:00 UTC #1127 - GitHub, accessed February 21, 2026, https://github.com/monero-project/meta/issues/1127
46. transcript.txt - Fireside.fm, accessed February 21, 2026, https://media24.fireside.fm/file/fireside-images-2024/podcasts/transcripts/9/9813d709-5a6d-4a87-8b5e-46ebf14f99d6/episodes/4/41b61f46-96b6-4efa-a75e-ae9af0539eb1/transcript.txt